SECTION 4 · SUBSECTION 4.1
END-TO-END LOGIC FLOW

4.1.1 INTRODUCTION.

A run is one traversal of the shared state object. The orchestrator resolves the repository address, asks GitHub for the default branch, then hands the same *ArchaeonState* to each agent in turn. Agents never import one another; they only read fields and write new ones. The only departure from sequence is the middle group: graph, complexity and retrieval are submitted to a three-thread pool, then joined in a fixed order so that graph statistics can be bound into the complexity scores before they are consumed (figure 4.1-1).

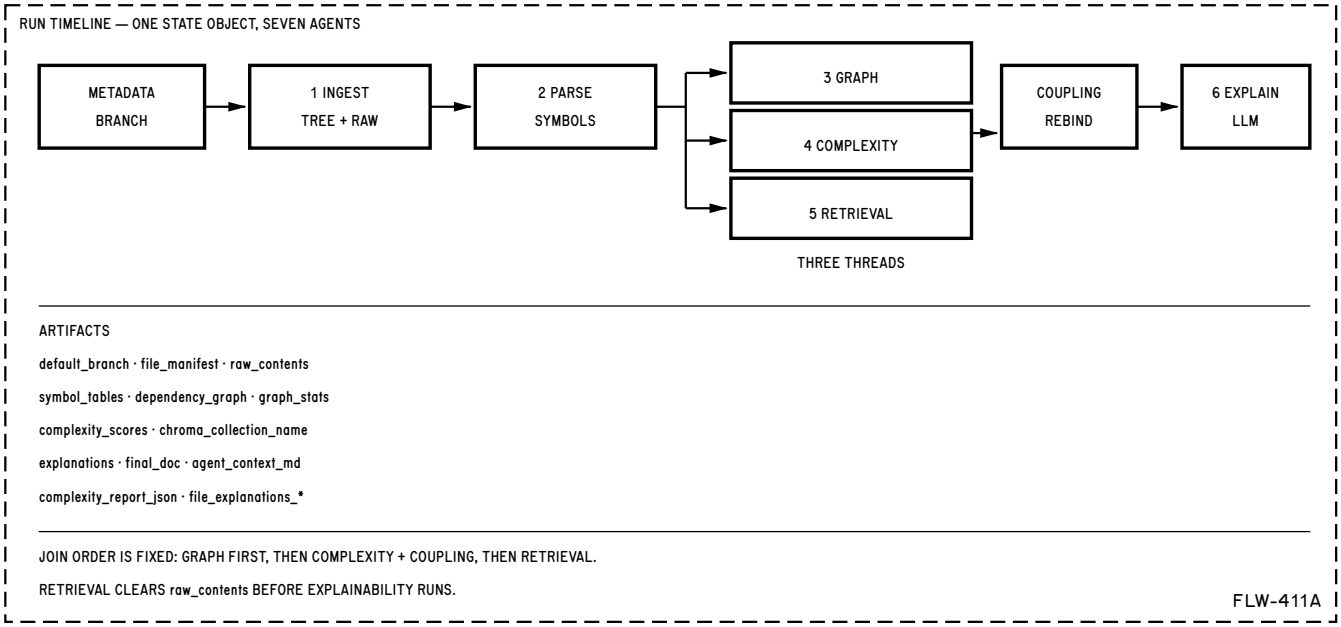


Figure 4.1-1. Phase Order, Fan-Out and State Artifacts

4.1.2 FUNCTIONAL DESCRIPTION.

Metadata resolution is a single *GET /repos/{owner}/{repo}*. Ingestion then lists the default branch recursively, filters paths by extension, excluded directory and size, and fetches raw text from *raw.githubusercontent.com*. The manifest is sorted by depth then path before any cap is applied. Parsing builds one symbol table per file, recording functions, classes, imports, docstrings and parse errors; a parse failure is data, not an abort.

The middle group starts with all three futures alive. The orchestrator waits for the graph, then for complexity, immediately calls *apply_graph_coupling* to replace placeholder coupling values with real in-degree and out-degree, and only then waits for retrieval. Retrieval chunks each parsed symbol, embeds the chunks in request batches and writes them to a per-job Chroma collection. After the join, explainability ranks files and asks the hosted model for prose; the document generator assembles four outputs and returns control.

4.1.2.1 CLI and API Divergence.

The CLI calls *run_pipeline* and then writes ten files itself, one save function at a time. The API calls the same function inside *pipeline_runner*, serialises a subset of state into one JSON record, and never writes files. The API result therefore omits nothing the browser needs but does omit *symbol_tables* details, and the CLI writes *file_explanations* artifacts that the API returns only inside the aggregate result.

4.1.3 PERFORMANCE AND DESIGN DATA.

Table 4.1-1. Phase Inputs, Outputs and Limits

Phase	Reads	Writes
Metadata	owner, repo, token	default_branch
1 Ingestion	URL, branch	manifest, raw text
2 Parse	raw text, paths	symbol tables
3 Graph	imports, manifest	graph, cycles, order
4 Complexity	raw text, symbols	scores, risk
5 Retrieval	raw text, symbols	collection, chunks
Coupling	graph stats	coupling, final risk
6 Explain	scores, graph, Chroma	explanations
7 Document	all of the above	four documents

NOTE

Retrieval empties *raw_contents* on success to release memory. Explainability's fallback path then finds no raw text and must rely on Chroma chunks, which exist. If retrieval stored zero chunks it returns before clearing, so the fallback still has text but the collection is empty.

FLOW CHECK	REVIEWER
1 TRACE - manifest count to parse count	
2 CONFIRM - graph waited before coupling	
√ JOIN - retrieval after coupling	
3 CHECK - explanations reference real files	
4 VERIFY - document headings not empty	

FLW